Predictive Advantage, Regime Dependence, and Direction-Specific Causal Control in Kinetic Ising Dynamics

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Abstract

We study how the choice of description affects predictive and causal information about future evolution in the two-dimensional kinetic Ising model. We report four empirical results together with a predictor-handle dissociation result. Fine descriptions based on locally exposed and junction-type defects outperform a block-variance coarse rival for predicting future domain-wall change in the ordered phase; the best organism is $E_{\geq 3} + J$, with predictive advantage $\Delta R^2 = +0.415$ (95% CI [+0.389, +0.443]). (**Axis 2, regime crossover**) This advantage reverses sign in the disordered phase (+0.183 ordered, −0.064 disordered), showing that the fine-over-coarse advantage is regime-dependent. (**Axis 3, target specificity**) Under a fine-over-coarse ΔR^2 criterion, future domain-wall change ΔW is the target most specifically favoured by the fine description ($\Delta R^2 = +0.415$); this is not merely the most predictable target in absolute terms. (**Axis 4, causal control**) Biasing the Glauber generator toward wall-reducing flips at exposed sites reduces future wall loss, while biasing against the same flips increases it; this direction-specific sign flip holds across all tested perturbation strengths (4/4). (**Dissociation**) The best predictor of ΔW—the coarse macro-state $[W_0, B_{\mathrm{mag}}]$ ($R^2 = 0.378$)—differs from the causal handle family (exposed-site structure, $R^2 = 0.181$), with gap +0.197. The best predictor and the best causal handle are empirically distinct within the same system at the same temperature.

1 Introduction

A common working assumption in statistical physics is that different descriptions of the same system may differ in predictive usefulness—but which description is best, and whether that ranking is stable across regimes and tasks, is not always obvious. This paper studies how predictive and causal usefulness depend on the choice of description in a simple, well-understood dynamical system.

We ask which fine description of an ordered Ising configuration predicts future interface evolution best; whether that advantage is regime-dependent; which future target is most specifically favoured by the fine description relative to a coarse rival; whether the fine structural description provides a direction-specific causal handle; and whether the best predictor coincides with the best causal handle. The first question fixes the target (ΔW) and asks which organism wins; the third question fixes the champion organism and asks which target it most specifically favours over the coarse rival—these are distinct. The answers are, respectively: $E_{\geq 3} + J$; yes (reverses in the disordered phase); ΔW; yes (4/4 sign-flip); and no (predictor and handle are dissociable).

Why the Ising model. The 2D kinetic Ising model with Glauber dynamics (Glauber, 1963; Ising, 1925) is one of the few systems for which domain-wall coarsening, defect densities, and critical exponents are all precisely characterised (Bray, 1994). Domain walls provide a structurally meaningful target family: interfaces are the elementary reorganisation objects under coarsening, and their evolution is driven by specific local geometric features of the current configuration. An isolated minority spin surrounded by four unlike neighbours—a site with exposure degree $d_c = 4$—contributes four domain-wall bonds, is energetically costly, and is a natural structural precursor of upcoming interface simplification. This geometry makes the Ising model a clean and interpretable testbed.

Why description choice matters. Finer descriptions always contain more bits, but more bits do not automatically produce better predictions for a specific target. The predictive signal for interface-level change lies in sparse, geometrically special objects—exposed tips, junction bends—that coarse block averages discard. In the disordered phase no such sparse structure exists; coarse summaries recover the predictive advantage. And predictive sufficiency does not confer causal access: as we demonstrate, the variable that best forecasts ΔW is not the variable whose structure can be used to controllably change ΔW.

Paper organisation. This paper establishes four empirical results and one dissociation result: **(1)** organism comparison in the ordered phase; **(2)** regime crossover demonstrating regime-dependence; **(3)** target specificity under a fine-over-coarse criterion; **(4)** direction-specific causal control via generator alignment; and **(D)** predictor-handle dissociation. Section 2 reviews related work. Section 3 presents the model, observables, and causal protocol. Section 4 reports results with figures and tables. Sections 5–6 discuss implications and conclude.

2 Related Background

Coarse-graining and renormalisation. The renormalisation group (Kadanoff, 1966; Wilson, 1975) is the canonical framework for relating descriptions at different scales. In that tradition, coarse-graining maps fine configurations to block variables, and RG fixed points characterise universality classes. Our approach is complementary: we ask empirically, for a given target and horizon, which description achieves higher out-of-sample predictive accuracy. Israeli and Goldenfeld (2006) showed that cellular automaton coarse-graining can produce exact reduced models, illustrating the system-specificity of which description is "best."

Prediction versus causation. The distinction between prediction and intervention is formalised in the do-calculus of Pearl (2009) and the interventionist philosophy of Woodward (2003). A variable may be a strong predictor while being causally inert if it is correlated with the true causal driver. We demonstrate this dissociation empirically within Ising: the best predictor of ΔW does not coincide with the structure that can be used to controllably change ΔW.

Structural defects in coarsening. The role of domain-wall defects in Ising coarsening is well studied (Bray, 1994; Spirin et al., 2001). Isolated minority spins and junction-type sites are energetically unstable and drive local coarsening events. We quantify how well such defects predict future macroscopic wall change and test whether they serve as intervention targets.

Macro-micro comparisons in prediction and causation. The causal emergence programme (Hoel et al., 2013, 2017) asks whether macro descriptions can have more causal organisation than micro ones, measured by effective information. Our approach differs: we use standard regression criteria for predictive accuracy and direct intervention experiments for causal leverage, without assuming any universal scale ranking.

3 Model and Methods

3.1 The 2D kinetic Ising model

We study the ferromagnetic Ising model on a 64×64 square lattice with periodic boundary conditions. Each site i carries spin $\sigma_i \in \{-1, +1\}$. Dynamics are governed by single-spin-flip Glauber updates (Glauber, 1963):

$$p(\sigma_i \to -\sigma_i) = \frac{1}{1 + \exp\big(2\beta h_i \sigma_i\big)}, \tag{1}$$

where $h_i = \sum_{\langle j \rangle} \sigma_j$ and $\beta = 1/T$. A checkerboard update scheme avoids spatial bias. The 2D Ising critical temperature is $T_c \approx 2.269$.

Two anchor temperatures are used: ordered $T = 1.80$ (well inside the ferromagnetic phase) and disordered $T = 2.50$ (above T_c). Each configuration is equilibrated for 1000 sweeps before measurement. Predictive targets are measured at horizon $K = 20$ sweeps.

3.2 Structural observables

Domain-wall count. $W(\sigma) = \sum_{\langle i,j \rangle} \mathbf{1}[\sigma_i \neq \sigma_j]$. Future change: $\Delta W = W(t_0 + K) - W(t_0)$.

Exposed-site counts. Exposure degree: $d_c(i) = \sum_{\langle j \rangle} \mathbf{1}[\sigma_j \neq \sigma_i]$.

$$E(\sigma) = \sum_i \mathbf{1}[d_c(i) = 4] \quad \text{(isolated minority spins)}, \tag{2}$$

$$E_{\geq 3}(\sigma) = \sum_i \mathbf{1}[d_c(i) \geq 3] \quad \text{(highly exposed sites)}. \tag{3}$$

Junction count. $J(\sigma) = \sum_i \mathbf{1}[d_c(i) = 2 \text{ and non-collinear}]$. Junctions mark the bends and corners of wall segments.

Composite organisms. $E_{\geq 3} + J$ is the two-column matrix $[E_{\geq 3}, J]$; each column receives a separate regression coefficient. $E + J$ uses $[E, J]$. These are not scalar sums.

Coarse rival.

$$\phi_{\text{var8}}(\sigma) = \frac{1}{n_b} \sum_{b=1}^{n_b} \left(|\bar{\sigma}_b| - \tfrac{1}{n_b} \sum_{b'} |\bar{\sigma}_{b'}| \right)^2, \tag{4}$$

the variance of block-averaged order parameter across 8×8 blocks.

3.3 Predictive evaluation

Ridge regression, 5-fold cross-validation, $N = 3000$ configurations. Primary statistic:

$$\Delta R^2 = R^2(\text{fine organism}) - R^2(\phi_{\text{var8}}). \tag{5}$$

Confidence intervals use pairwise bootstrap resampling ($B = 1000$) with the same configuration set for fine and coarse predictions.

3.4 Causal protocol: aligned and misaligned generator bias

Rather than forcing state changes at t_0, we modify the Glauber transition probability throughout the K-step evolution. At each exposed site i ($d_c(i) = 4$), the local wall-change contribution of flipping σ_i is

$$\delta W_i = \sum_{\langle j \rangle} \big[\mathbf{1}[\sigma_j \neq -\sigma_i] - \mathbf{1}[\sigma_j \neq \sigma_i] \big]. \tag{6}$$

For a fully exposed site, $\delta W_i < 0$.

Two arms are defined:

Aligned: $\delta W_i < 0$ transitions promoted ($p \to p + \varepsilon$); $\delta W_i > 0$ suppressed ($p \to p - \varepsilon$).

Misaligned: Opposite: $\delta W_i < 0$ suppressed, $\delta W_i > 0$ promoted.

Both share initial configuration $\sigma(t_0)$ and are compared to an unperturbed baseline arm. Primary output:

$$\text{gap} = \Delta W_{\text{baseline}} - \Delta W_{\text{arm}}, \tag{7}$$

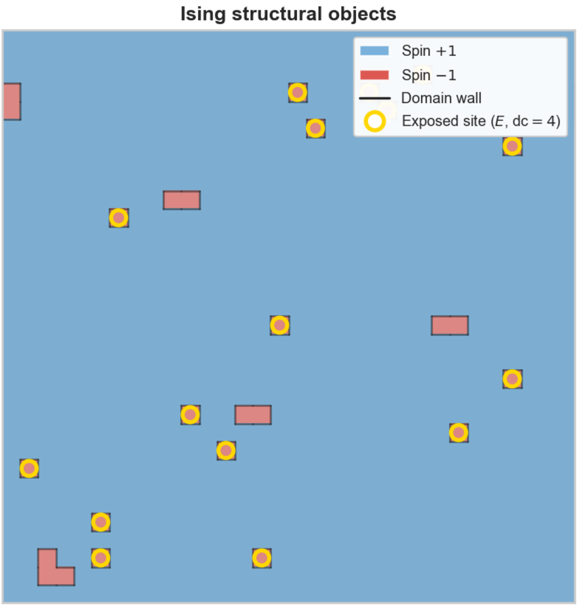

Figure 1: **Structural objects in an ordered Ising configuration.** 32×32 patch at $T = 1.80$. Red/blue: spin $-1/+1$ domains. Black lines: domain walls. Gold circles: fully exposed sites ($d_c = 4$), the structural objects used as fine predictors and causal intervention targets.

positive meaning the arm loses fewer walls. The *sign-flip discriminator* requires aligned gap > 0 and misaligned gap < 0, both with CIs excluding zero.

4 Results

4.1 Structural objects and intuition

Figure 1 illustrates the key objects. Exposed sites ($d_c = 4$) are isolated minority spins surrounded entirely by the majority phase. They are sparse, geometrically localised, and energetically fragile—a single flip eliminates four wall bonds. Junction sites ($d_c = 2$, corner type) mark the bends of wall segments. Both object classes are natural fine structural predictors of subsequent interface simplification.

4.2 Axis 1: organism comparison

Figure 2 shows fine-over-coarse predictive advantages for all fine organisms at $T = 1.80$, target ΔW.

The best organism is E_ge3_plus_J, achieving $\Delta R^2 = +0.415$ (CI [+0.389, +0.443]). Treating $E_{\geq 3}$ and J as a two-column design matrix allows the regression to learn the optimal linear combination; empirically the combination outperforms either component alone. The coarse rival ϕ_{var8} measures large-scale domain ordering, which changes slowly in the ordered phase; the predictive signal for interface-level change lies precisely in the fine-scale topology that the block average discards.

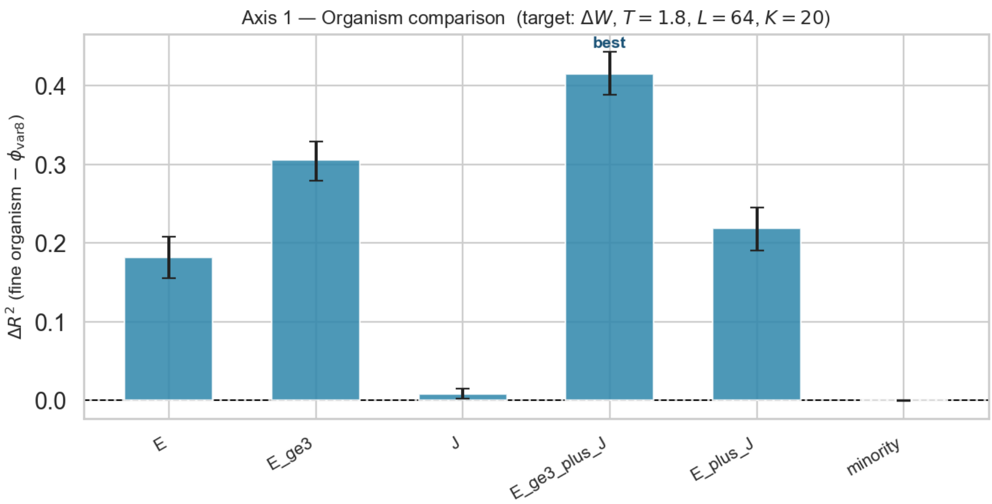

Figure 2: **Organism comparison ($T = 1.80$, target ΔW).** $\Delta R^2 = R^2(\text{fine}) - R^2(\phi_{\text{var8}})$ with bootstrap 95% CIs. Blue: fine wins; a red bar would indicate coarse wins. $E_{\geq 3} + J$ (two-column model) achieves the largest advantage.

4.3 Axis 2: regime crossover

Table 1: **Regime crossover: ΔR^2 for E vs ϕ_{var8} on ΔW.** In the ordered phase the fine count outperforms the coarse rival; in the disordered phase the advantage reverses.

Regime	Temperature	ΔR^2	95% CI
Ordered	$T = 1.80$	$+0.183$	$[+0.158, +0.209]$
Disordered	$T = 2.50$	-0.064	$[-0.084, -0.043]$

Table 1 shows the regime comparison for E versus ϕ_{var8}. Fine wins in the ordered phase ($+0.183$); coarse wins in the disordered phase (-0.064). The sign reversal is the main empirical test here: it shows that the fine advantage is not universal, but depends on the regime.

The physical interpretation is straightforward. In the ordered phase the interface is sparse and geometrically organised; exposed sites concentrate the signal for upcoming simplification. In the disordered phase almost every site qualifies as "exposed," the concept loses discriminative power, and large-scale bulk fluctuations captured by ϕ_{var8} become the better predictor. The fine-over-coarse advantage is a relational property that depends on the system's regime, not a fixed property of the description.

4.4 Axis 3: target specificity

Figure 3 shows which future target is most specifically favoured by E_ge3_plus_J relative to ϕ_{var8}. The criterion is ΔR^2, not raw R^2. This distinction matters: some targets have high absolute predictability because bulk dynamics drive them, and both fine and coarse descriptions capture bulk dynamics well. Measuring fine-over-coarse ΔR^2 identifies the

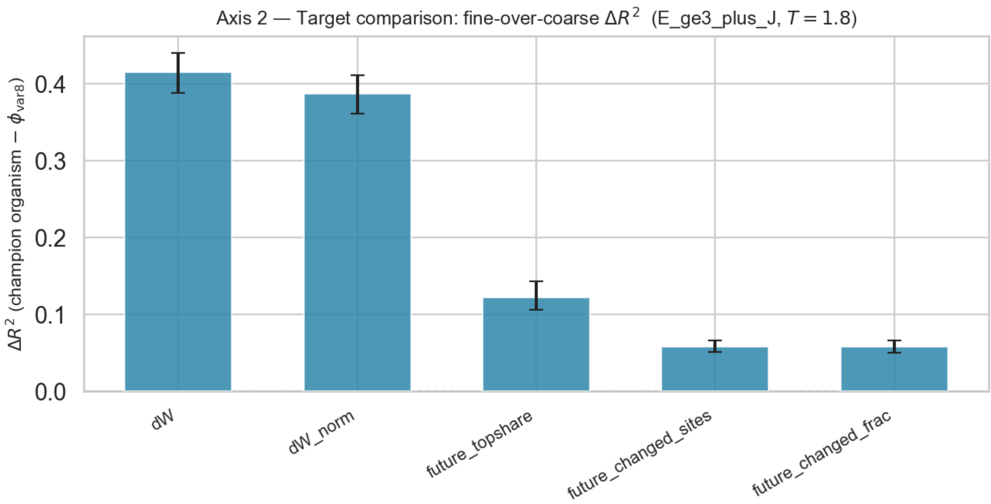

Figure 3: **Target comparison ($T = 1.80$, predictor E_ge3_plus_J).** ΔR^2 = fine − coarse across future targets. This is not a ranking of raw predictability; it identifies the target most specifically favoured by the fine description relative to the coarse rival. ΔW achieves the largest fine advantage.

target where the *marginal value* of fine over coarse information is largest.

The answer is ΔW: $\Delta R^2 = +0.415$, fine $R^2 = 0.413$, coarse $R^2 = -0.004$. Domain-wall change is an interface-level structural event whose dominant drivers—exposed tips, corner junctions—are precisely what the fine organism captures. The central result of this axis: **ΔW is the target most specifically favoured by fine structural descriptions in ordered Ising, under a fine-over-coarse criterion.**

4.5 Axis 4: direction-specific causal control

Table 2: **Axis 4 sign-flip ladder.** Gap = baseline − arm on ΔW. Positive: arm loses fewer walls. Negative: arm loses more. 4/4 sign flips.

ε	Aligned gap	Misaligned gap	Sign flip
0.02	> 0, CI excl. 0	< 0, CI excl. 0	✓
0.05	> 0, CI excl. 0	< 0, CI excl. 0	✓
0.10	$+5.75$, CI excl. 0	-26.68, CI excl. 0	✓
0.20	> 0, CI excl. 0	< 0, CI excl. 0	✓

Figures 4–5 and Table 2 present the causal generator test. The result is EARNED: all four ε values pass the sign-flip discriminator. The aligned arm consistently loses fewer walls than baseline (gap at $\varepsilon = 0.10$: $+5.75$, CI excludes zero); the misaligned arm consistently loses more (gap: -26.68, CI excludes zero).

Why sign flip $\neq$ generic perturbation. Comparing one biased arm to the unperturbed baseline would only show that a modified kernel differs from the unmodified one—a trivially

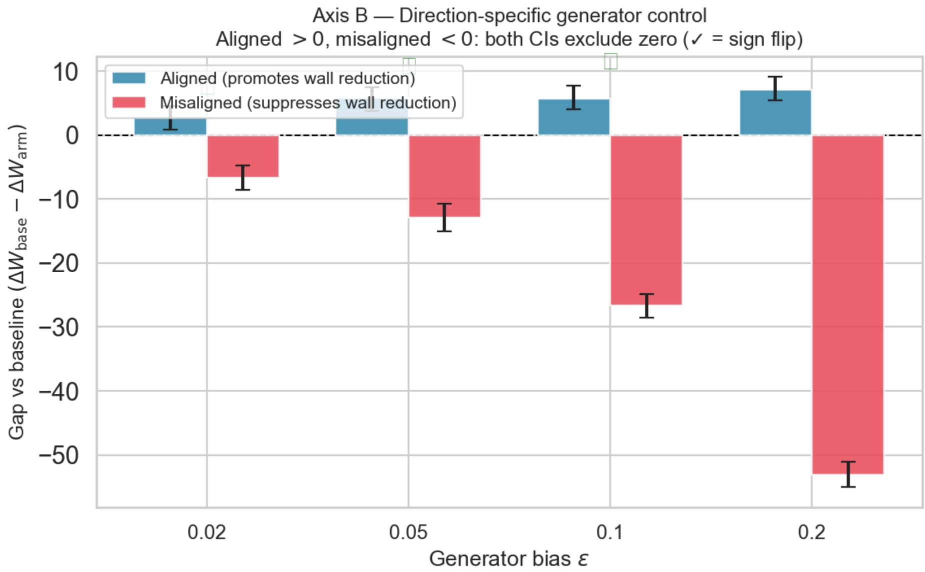

Figure 4: **Direction-specific generator causal test.** Bars: gap (baseline − arm) on ΔW. Blue (aligned): promoting wall-reducing flips at exposed sites reduces future wall loss. Red (misaligned): suppressing those flips increases wall loss. All four ε values produce a sign flip (✓): aligned CI > 0, misaligned CI < 0, both excluding zero. The sign reversal supports the interpretation that the effect is structurally specific, rather than a generic perturbation effect.

true result. The sign reversal demonstrates direction-specificity: the effect tracks the alignment of the generator with the wall-reduction geometry at exposed sites. Promoting wall-reducing transitions at those sites reduces future wall loss; suppressing them increases it. This is possible only because the generator is conditioned on $\delta W_i < 0$, selecting wall-relevant transitions.

Target specificity. Figure 5 shows the ΔM effect. The dominant causal effect is on ΔW. ΔM coupling is comparatively small at low and high ε. We do not claim ΔM is strictly null across all ε; we claim the intervention is primarily directed at domain-wall evolution, consistent with its design.

Scope. Axis 4 earns: in ordered Ising at $T = 1.80$, a structure-aligned generator perturbation produces a direction-specific causal effect on future ΔW. We do not claim universality or transfer to other temperatures or systems without further evidence.

4.6 Predictor-handle dissociation

The best predictor of future ΔW is the coarse macro-state $[W_0, B_{\text{mag}}]$—not the fine exposed-site family:

$$R^2([W_0, B_{\text{mag}}]) = 0.378, \qquad R^2(E) = 0.181, \qquad \text{gap} = +0.197. \tag{8}$$

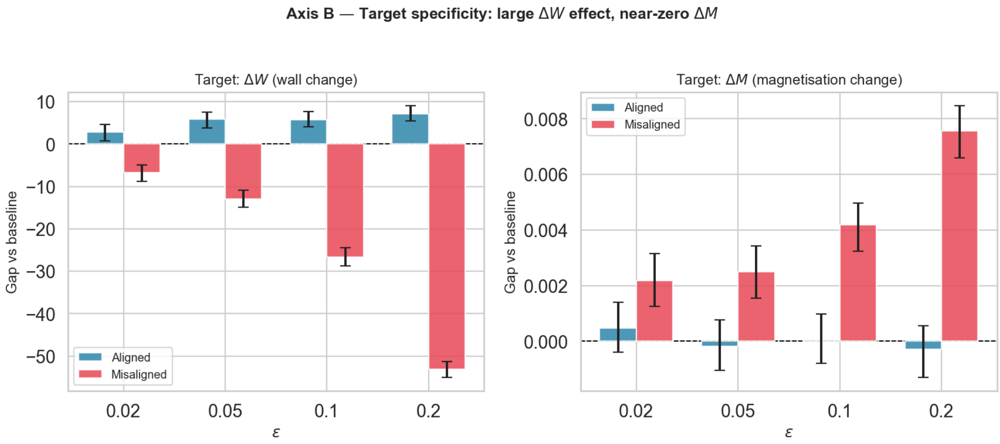

Figure 5: **Target specificity: ΔW versus ΔM.** Left: the ΔW sign flip. Right: corresponding ΔM effect. The dominant effect is on ΔW. ΔM coupling is comparatively small at low and high ε; a modest drift appears at intermediate ε but does not approach the magnitude of the ΔW effect.

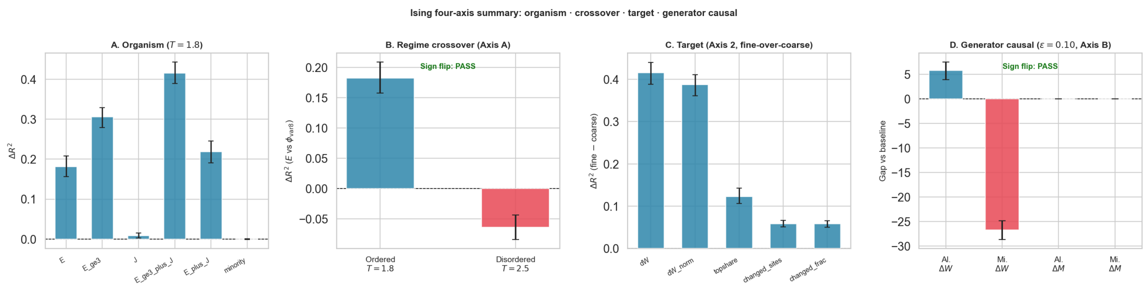

Figure 6: **Four-component summary panel.** (**A**) Axis 1 organism comparison: $E_{\geq 3} + J$ is the champion. (**B**) Axis 2 regime crossover: fine wins ordered, coarse wins disordered. (**C**) Axis 3 target comparison: ΔW is the most fine-specific target. (**D**) Axis 4 causal test at $\varepsilon = 0.10$: aligned gap on ΔW positive, misaligned negative; ΔM comparatively small. $N = 3000$ predictive worlds, 3000 causal worlds, $L = 64$, $K = 20$.

The predictor-superiority gap exceeds 0.08.

This creates a sharp dissociation: the fine organism E (and $E_{\geq 3} + J$) provides causal leverage over ΔW (Axis 4), while the coarse macro-state $[W_0, B_{\text{mag}}]$ better predicts the same quantity. **The best predictor and the best causal handle are different objects in the same system at the same temperature.**

Two caveats. First, this does not mean coarse descriptions are generally better: $[W_0, B_{\text{mag}}]$ is predictively sufficient for ΔW on a statistical forecasting task, but it does not identify the specific sites that drive coarsening. The fine structure does identify those sites, and that is precisely what enables the generator-level intervention of Axis 4. Second, the dissociation is a property of this specific predictor-target pair at this temperature; it may not hold for other pairs.

4.7 Synthesis

Figure 6 summarises all four components. In the ordered phase, exposed-site and junction structure carries predictive information that coarse averaging discards; this advantage is regime-specific. Under the fine-over-coarse criterion, ΔW is the most fine-specific target. The causal sign-flip confirms that the structural handle is direction-specific. And the predictor-handle dissociation shows that predictive sufficiency and causal access are different properties of a description. All four results are independently earned and jointly coherent.

5 Discussion

5.1 What the ordered-regime fine advantage means

The fine advantage is not a trivial consequence of resolution. More bits do not automatically improve out-of-sample prediction for a specific target. What $E_{\geq 3} + J$ captures is the fragility topology of the interface: the geometrically special locations most likely to simplify in the next K steps. Coarse 8×8 block averages wash out this topology. The result is consistent with a broader principle: fine-description advantage appears when the predictive signal is concentrated in sparse, structurally special objects that coarse summaries discard—and disappears when those objects lose their sparse, special character.

5.2 Why the regime crossover is the main empirical test

The sign reversal between ordered and disordered phases rules out three confounds simultaneously. It rules out that fine organisms are always better because they contain more information. It rules out that the advantage is an artefact of the specific coarse rival chosen. And it provides a physical narrative tied to the phase structure: as thermal fluctuations grow near T_c, interfaces roughen, the exposed-site concept loses discriminative power (Bray, 1994), and bulk fluctuations become the better predictor. A finding without a crossover would be scientifically weaker.

5.3 Why Axis 3 requires the fine-over-coarse criterion

Measuring Axis 3 by raw R^2 would conflate fine-over-coarse advantage with raw predictability. Some targets are easy to predict in absolute terms because bulk dynamics drive them, and both fine and coarse descriptions track bulk dynamics well. The fine-over-coarse ΔR^2 criterion identifies the target where the *marginal value* of fine over coarse information is largest. That target is ΔW, consistently and for clear structural reasons. Reporting Axis 3 by raw R^2 would risk pointing to the wrong target and would weaken the structural argument.

5.4 Why Axis 4 is stronger than a generic gap

Any modification to the Glauber kernel differs from the unmodified kernel. What Axis 4 demonstrates is that the *direction* of the modification matters, in the way predicted by the fine structural description. The misaligned arm provides the internal negative control. Reversing the alignment reverses the causal effect. This pattern supports the interpretation that the effect is structurally specific, rather than a generic perturbation effect (Pearl, 2009; Woodward, 2003).

5.5 The dissociation and its implications

The predictor-handle dissociation (Section 4.6) is conceptually the most significant result. It shows that two roles—best predictor of ΔW and best causal handle for ΔW—are filled by different variables in the same system at the same temperature. The coarse macro-state $[W_0, B_{\text{mag}}]$ predicts well because it summarises how much coarsening remains possible; it does not identify which sites will drive it. The fine exposed-site structure identifies those sites; that identification is what enables Axis 4. A scientist relying on predictive models alone would identify the wrong intervention target.

We emphasise: the dissociation does not mean coarse descriptions are generally better. It means that predictive sufficiency and causal accessibility are different properties of a description, and they can be empirically separated.

5.6 Limitations and extensions

Results are scoped to ordered 2D Ising, $L = 64$, $K = 20$. Transfer to other models is not automatic. Preliminary results from Potts ($q = 3$) and Clock ($q = 6$) suggest the fine-over-coarse predictive advantage transfers in the ordered phase, but the Axis 4 causal result is topology-sensitive: the binary-spin structure is required for a single flip to eliminate all four domain-wall bonds simultaneously. In multi-colour Potts, no tested object fills that role. Extension to quantum systems (Bose-Hubbard chain) and continuous-field systems (XY, ϕ^4) requires adapting the exposed-defect concept and is the subject of companion work.

6 Conclusion

We have reported four empirical results and one dissociation result for fine and coarse descriptions of ordered Ising dynamics.

1. **Axis 1 (organism comparison):** $E_{\geq 3}+J$ achieves $\Delta R^2 = +0.415$ (CI $[+0.389, +0.443]$) for future ΔW in the ordered phase.

2. **Axis 2 (regime crossover):** The fine advantage reverses: $+0.183$ (ordered) versus -0.064 (disordered). The fine-over-coarse advantage is regime-dependent.

3. **Axis 3 (target specificity):** Under the fine-over-coarse criterion, ΔW is the most fine-specific target (fine $R^2 = 0.413$, coarse $R^2 = -0.004$, $\Delta R^2 = +0.415$).

4. **Axis 4 (direction-specific causal control):** Structure-aligned generator bias at exposed sites produces a 4/4 sign-flip on future ΔW. All CIs exclude zero.

5. **Predictor-handle dissociation:** $R^2([W_0, B_{\text{mag}}]) = 0.378$, $R^2(E) = 0.181$, gap $= +0.197$. The best predictor and the best causal handle are different objects.

In this system, the relative usefulness of fine and coarse descriptions depends on the target and the regime. The Ising model provides a clean test case for these distinctions.

Data and Code Availability

All code available at https://github.com/kunalb541/Papers_Git

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